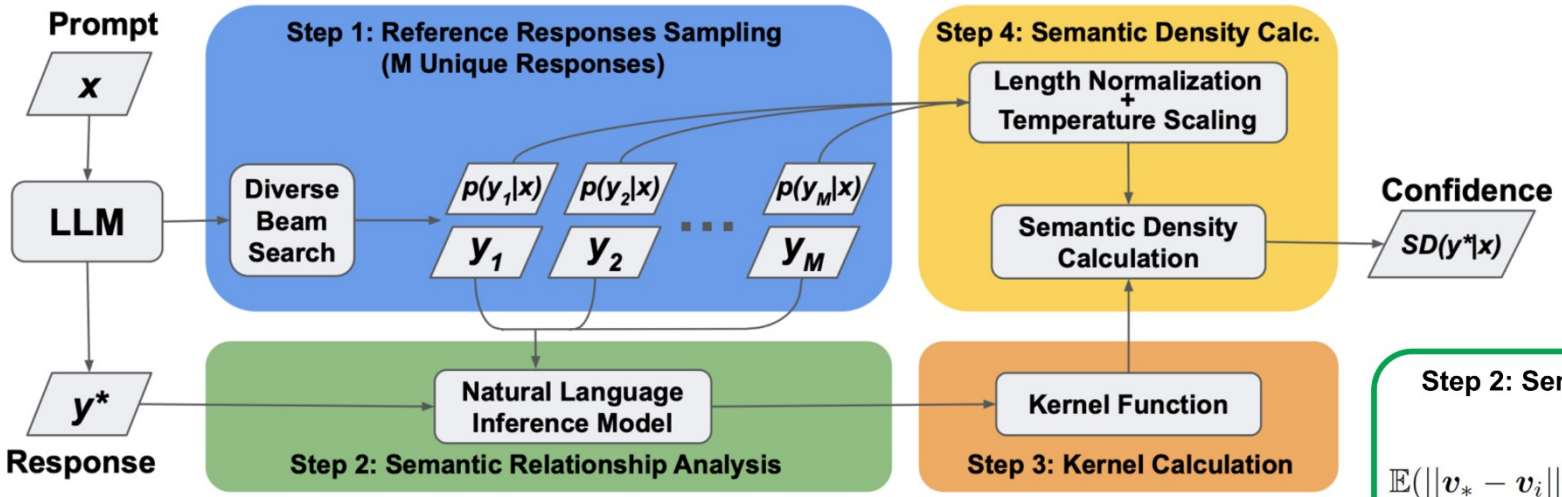
Semantic Density

Semantic Density[79]



Algorithm 1 Procedure for deploying semantic density

Require:

$\boldsymbol{y}_*$: target response that needs confidence measurement

$\boldsymbol{x}$: original prompt for generating $\boldsymbol{y}_*$

M: number of unique reference responses to be sampled given $\boldsymbol{x}$

Ensure:

$\mathrm{SD}(\boldsymbol{y}_*|\boldsymbol{x})$: semantic density for $\boldsymbol{y}_*$ given $\boldsymbol{x}$

Step 1: Reference Response Sampling:

1: sample M unique reference responses $\boldsymbol{y}_i$ (for $i = 1, 2, \cdots, M$) with prompt $\boldsymbol{x}$ on the original LLM using diverse beam search, and record each corresponding length-normalized sampling probability $\sqrt[L_i]{p(\boldsymbol{y}_i|\boldsymbol{x})}$

Step 2: Semantic Relationship Analysis:

2: **for** $i = 1$ to M **do**

3: obtain $p_{\mathrm{c}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x})$ and $p_{\mathrm{n}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x})$ using NLI classification model

4: calculate expectation $\mathbb{E}(||\boldsymbol{v}_* - \boldsymbol{v}_i||^2) = p_{\mathrm{c}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x}) + \frac{1}{2} \cdot p_{\mathrm{n}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x})$

Step 3: Kernel Function Calculation:

5: **for** $i = 1$ to M **do**

6: calculate kernel function value using the expectation of $||\boldsymbol{v}_* - \boldsymbol{v}_i||^2$, given by:

$K(\boldsymbol{v}_* - \boldsymbol{v}_i) = (1 - \mathbb{E}(||\boldsymbol{v}_* - \boldsymbol{v}_i||^2))\mathbf{1}_{\mathbb{E}(||\boldsymbol{v}_* - \boldsymbol{v}_i||) \leq 1}$

Step 4: Semantic Density Calculation:

7: calculate semantic density: $\mathrm{SD}(\boldsymbol{y}_*|\boldsymbol{x}) = \frac{1}{\sum_{i=1}^{M} \sqrt[L_i]{p(\boldsymbol{y}_i|\boldsymbol{x})}} \sum_{i=1}^{M} \sqrt[L_i]{p(\boldsymbol{y}_i|\boldsymbol{x})} K(\boldsymbol{v}_* - \boldsymbol{v}_i)$

Step 2: Semantic Relationship Analysis

Contradictory Prob. Neutral Prob. Entailment Prob.

$$\mathbb{E}(||\boldsymbol{v}_* - \boldsymbol{v}_i||^2) = 1^2 \cdot p_{\mathrm{c}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x}) + (\tfrac{\sqrt{2}}{2})^2 \cdot p_{\mathrm{n}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x}) + 0^2 \cdot p_{\mathrm{e}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x})$$
$$= p_{\mathrm{c}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x}) + \tfrac{1}{2} \cdot p_{\mathrm{n}}(\boldsymbol{y}_*, \boldsymbol{y}_i|\boldsymbol{x})$$

Step 3: Kernel Calculation

$$K(\boldsymbol{v}_* - \boldsymbol{v}_i) = (1 - ||\boldsymbol{v}_* - \boldsymbol{v}_i||^2)\mathbf{1}_{||\boldsymbol{v}_* - \boldsymbol{v}_i|| \leq 1}$$

Estimated by $\mathbb{E}(||\boldsymbol{v}_* - \boldsymbol{v}_i||^2)$

Step 4: Semantic Density Calc.

$$\mathrm{SD}(\boldsymbol{y}_*|\boldsymbol{x}) = \frac{1}{\sum_{i=1}^{M} p(\boldsymbol{y}_i|\boldsymbol{x})} \sum_{i=1}^{M} p(\boldsymbol{y}_i|\boldsymbol{x}) K(\boldsymbol{v}_* - \boldsymbol{v}_i)$$

Post-processing by Length-normalization + Temperature Scaling

Sampling Probability by LLM

[79] Semantic Density: Uncertainty Quantification for Large Language Models through Confidence Measurement in Semantic Space, Neurips 2024

Semantic Density[79]

- **Task Type:** Open-ended generation

- **Introduction:** Semantic Density measures LLM uncertainty by evaluating how tightly plausible alternative responses cluster around the primary answer in semantic space. It generates a main answer and diverse references, analyzes their relations via NLI, and computes a density score reflecting reasoning consistency. Semantic Density provides an interpretable uncertainty signal complementary to probability-based metrics.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding/Token-level	White-box	Post-hoc	Medium	Ranking

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